$$\frac{\partial w}{\partial s} = y^2z^32 + 2xyz^33 + 3xy^2z^2 = 2(2r + 3s + t)^2(3r + s + t)^3$$
$$+ 6(r + 2s + 3t)(2r + 3s + t)(3r + s + t)^3$$
$$+ 3(r + 2s + 3t)(2r + 3s + t)^2(3r + s + t)^2.$$

We now turn to formula (VI), i.e.

$$\frac{dy}{dx} = \left(\frac{dx}{dy}\right)^{-1}.$$

As we know from § 12.9, this formula concerns the derivative of an *inverse function.* Assuming that f has an inverse function f^{-1}, then

$$f \circ f^{-1}(x) = x$$

(exercise 7.16(5)). If f is differentiable at ξ and f^{-1} is differentiable at $\eta = f(\xi)$, then the formula (V) yields that

$$f'(\xi)(f^{-1})'(\eta) = I$$

from which formula (VI) follows.

When considering the real case we also gave conditions under which a suitable differentiable inverse exists. It follows from theorem 12.10 that it is sufficient in the real case if f' exists and is positive on some open interval containing ξ. The same is, of course, true if the word positive is replaced by the word negative. In the particular case when f' is continuous at ξ, it is therefore enough if $f'(\xi) \neq 0$.

A similar result holds in the vector case. Note that $f'(\xi) \neq 0$ in the real case corresponds to the requirement that the affine function tangent to f at ξ has an inverse function. The condition in the vector case which corresponds to this requirement is that the $n \times n$ matrix $f'(\xi)$ be *non-singular*, i.e.

$$\det f'(\xi) \neq 0.$$

We call the determinant $\det f'(\xi)$ the *Jacobian* of f at ξ. It is an important result that a suitable differentiable inverse of f exists provided that all the first order partial derivatives of f are continuous at ξ and $\det f'(\xi) \neq 0$. However, we shall neither explain the meaning of the word 'suitable' in this context nor offer a proof of the result. A proof would require apparatus that has not been developed in this book.

19.26 Exercise

(1) Explain why all functions f: $\mathbb{R}^n \to \mathbb{R}^m$ whose co-ordinates are all polynomials are differentiable at every point. What is the appropriate result when the co-ordinates are rational functions?

(2) Explain why each of the following functions is differentiable at the given point and compute its derivative there.

(i) $f\colon \mathbb{R}^2 \to \mathbb{R}$ at $(1, 2)$ where $f(x_1, x_2) = x_1^2 + 2x_1x_2 + x_2^3$.

(ii) $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ at $(\pi, 1)$ where

$$f(x_1, x_2) = \begin{pmatrix} \cos(x_1x_2) \\ \sin(x_1x_2) \end{pmatrix}.$$

(iii) $f\colon \mathbb{R}^2 \to \mathbb{R}^3$ at $(1, 1)$ where

$$f(x_1, x_2) = \begin{pmatrix} x_1 \\ x_1 + x_2 \\ x_1x_2 \end{pmatrix}.$$

(iv) $f\colon \mathbb{R}^3 \to \mathbb{R}^2$ at $(1, 2, 3)$ where

$$f(x_1, x_2, x_3) = \begin{pmatrix} x_1x_2^2 \\ x_2^2x_3^3 \end{pmatrix}.$$

(v) $f\colon \mathbb{R} \to \mathbb{R}^3$ at 0 where

$$f(x) = \begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix}$$

For each of these functions, write down the equations for the appropriate tangent flat.

(3) What is the slope (or rate of increase) of the function f of question 2(i) at the point $(1, 2)$ in the direction $(1, -1)$? What is the direction at $(1, 2)$ in which the slope is greatest? What is the value of the slope in this direction? Write down a vector which is normal to the contour $x_1^2 + 2x_1x_2 + x_2^3 = 13$ at $(1, 2)$. Also write down the equation of the tangent line at this point.

(4) Explain the circumstances under which formulae (I), (III) and (IV)′ of §19.21 make sense when L and M are matrices and x, y, z and c are column vectors. Prove that the formulae are valid when they make sense.

(5) Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ is differentiable everywhere and that the function $\phi\colon \mathbb{R}^2 \to \mathbb{R}$ is defined by $\phi(x_1, x_2) = f(x_1x_2)$. If $y = \phi(x_1, x_2)$, prove that

$$x_1 \frac{\partial y}{\partial x_1} = x_2 \frac{\partial y}{\partial x_2}$$

for all $(x_1, x_2) \in \mathbb{R}^2$ for which $f'(x_1x_2) \neq 0$.

(6) Suppose that f: $\mathbb{R}^2 \to \mathbb{R}$ is differentiable everywhere and that $y = f(x_1, x_2)$ where

$$x_1 = r\cos\theta$$

$$x_2 = r\sin\theta.$$

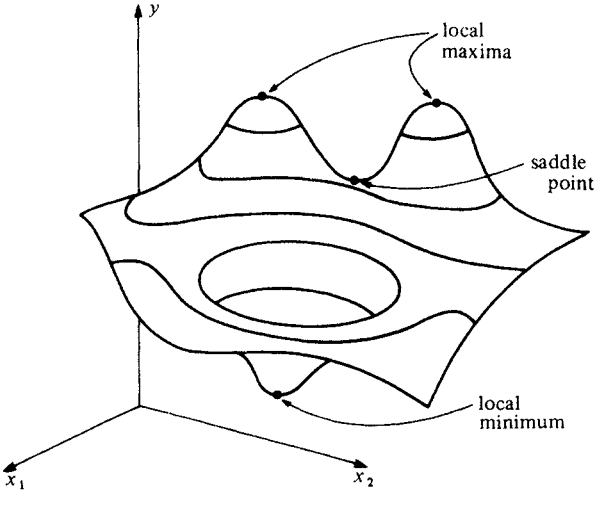
Express $\partial y/\partial x_2$ in terms of $\partial y/\partial r$ and $\partial y/\partial\theta$.

19.27 Stationary points

In §11.3 we defined a point ξ to be a *stationary point* of f if $f'(\xi) = 0$. We shall use precisely the same definition in the case of a real-valued function of a vector variable. In this case, of course, $f'(\xi)$ is a $1 \times n$ row vector and $f'(\xi) = 0$ means that all the co-ordinates of this row vector (i.e. the partial derivatives of f at ξ) are zero. In geometric terms, the fact that ξ is a stationary point means that the tangent hyperplane

$$y = f(\xi) + f'(\xi)(x - \xi)$$

reduces to $y = f(\xi)$, i.e. the tangent hyperplane is horizontal at a stationary point. The diagram shows some points on a surface $y = f(x_1, x_2)$ at which the tangent plane is horizontal.



Any local maximum or minimum at which f is differentiable is, of course, a stationary point. This follows immediately from corollary 19.9 and we quote the result as a theorem below. It is not at all uncommon, however, for stationary points in the vector case to correspond to *saddle points* as indicated in the